40th 1978-2017 Australian Mathematics Competition

AUSTRALIAN MATHEMATICS TRUST

AUSTRALIAN MATHEMATICS COMPETITION

AN ACTIVITY OF THE AUSTRALIAN MATHEMATICS TRUST

NAME _____

YEAR _____ TEACHER _____

2017
SIMPLIFIED CHINESE VERSION
UPPER PRIMARY DIVISION

AUSTRALIAN SCHOOL YEARS 5, 6 and PRIMARY 7*

*SOME STATES ONLY

TIME ALLOWED: 60 MINUTES

INSTRUCTIONS AND INFORMATION

GENERAL

1. Do not open the booklet until told to do so by your teacher.
2. You may use any teaching aids normally available in your classroom, such as MAB blocks, counters, currency, calculators, play money etc. You are allowed to work on scrap paper and teachers may explain the meaning of words in the paper. Mobile phones are not permitted.
3. Diagrams are NOT drawn to scale. They are intended only as aids.
4. There are 25 multiple-choice questions, each requiring a single answer, and 5 questions that require a whole number answer between 0 and 999. The questions generally get harder as you work through the paper. There is no penalty for an incorrect response.
5. This is a competition not a test; do not expect to answer all questions. You are only competing against your own year in your own country/Australian state so different years doing the same paper are not compared.
6. Read the instructions on the answer sheet carefully. Ensure your name, school name and school year are entered. It is your responsibility to correctly code your answer sheet.
7. When your teacher gives the signal, begin working on the problems.

THE ANSWER SHEET

1. Use only lead pencil.
2. Record your answers on the reverse of the answer sheet (not on the question paper) by FULLY colouring the circle matching your answer.
3. Your answer sheet will be scanned. The optical scanner will attempt to read all markings even if they are in the wrong places, so please be careful not to doodle or write anything extra on the answer sheet. If you want to change an answer or remove any marks, use a plastic eraser and be sure to remove all marks and smudges.

INTEGRITY OF THE COMPETITION

The AMT reserves the right to re-examine students before deciding whether to grant official status to their score.

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1-10 题，每题 3 分
Questions 1 to 10, 3 marks each

1. 请问比 17 大 20 的数是什么？

Which number is 20 more than 17?

(A) 3 (B) 27 (C) 37 (D) 217 (E) 2017

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2. 请问有多少个 200 g 的苹果派之总重量才等于 4 kg？

How many 200 g apple pies will weigh 4 kg?

(A) 2 (B) 20 (C) 50 (D) 80 (E) 200

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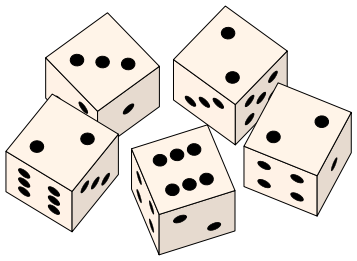
3. 投掷五枚骰子后的结果如图所示。
请问有几分之几的骰子顶面之点数为 2？

Five dice were rolled, and the results were as shown.
What fraction of the dice showed a two on top?



(A) $\frac{3}{4}$ (B) $\frac{1}{2}$ (C) $\frac{2}{3}$ (D) $\frac{2}{5}$ (E) $\frac{3}{5}$

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4. 小贞在户外用品店买了一个售价为 \$55 的背包与一个售价为 \$20 的指南针。
若她支付 \$100，请问她可以找回多少钱？

At the camping shop, Jane bought a rucksack for \$55 and a compass for \$20.
How much change did she get from \$100?

(A) \$25 (B) \$35 (C) \$45 (D) \$55 (E) \$65

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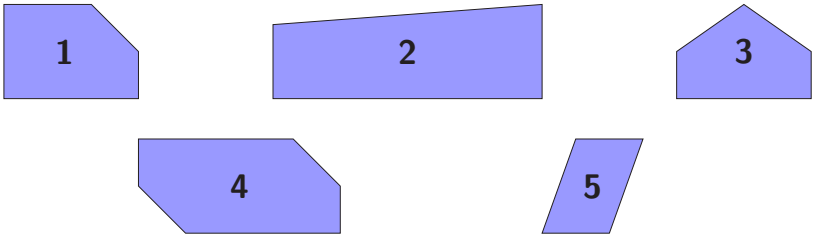
5. 请问下列哪些图案是五边形？



(A) 全部都是　(B) 只有图案 3　(C) 图案 3 与图案 4　(D) 图案 1 与图案 3　(E) 全部都不是

Which of these shapes are pentagons?

(A) all of the shapes　(B) shape 3 only　(C) shapes 3 and 4　(D) shapes 1 and 3　(E) none of the shapes

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6. 小蜜住在离学校 4 km 的地方；小娜住在与学校的距离恰为小蜜的住处与学校的距离之三倍，而小瓯的住处距离学校恰比小娜的住处距离学校还近 3 km。请问小瓯住在离学校多远的地方？

Mitchell lives 4 km from school. Naomi lives 3 times as far from school as Mitchell. Olivia lives 3 km closer to school than Naomi. How far does Olivia live from school?

(A) 9 km　(B) 3 km　(C) 15 km　(D) 13 km　(E) 21 km

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7. 小伦将一些数相加后所得到的和为 157。她发现她不小心将其中一个数 37 写成 73。请问正确的和应该是多少？

Helen is adding some numbers and gets the total 157. Then she realises that she has written one of the numbers as 73 rather than 37. What should the total be?

(A) 110　(B) 121　(C) 124　(D) 131　(E) 751

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8. 在公元 3017 年，某国造币厂回收所有硬币并发行新的硬币。

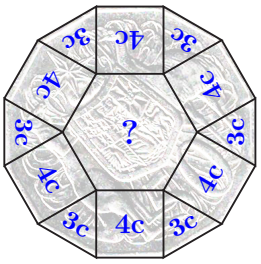
新的 50 元硬币被划分成六个三角形、六个正方形与一个六边形。每个三角形价值 3 元、每个正方形价值 4 元。请问这个六边形价值为多少元才能使得此硬币之总价值仍为 50 元？

(A) 3 元　(B) 8 元　(C) 18 元　(D) 20 元　(E) 43 元

In the year 3017, the Australian Mint recycled its coins to make new coins.

Each 50c coin was cut into six triangles, six squares, and one hexagon. The triangles were each worth 3c and the squares were each worth 4c.

How much should the value of the hexagon be to make the total still worth 50c?



(A) 3c　(B) 8c　(C) 18c　(D) 20c　(E) 43c

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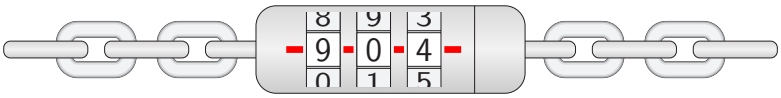
9. 小斐用如下图所示的号码锁来锁她的脚踏车，每道转盘上的数码都是从 0 到 9。

她每转动一次都将转盘往前或往后转动一个数码，这包括在 9 与 0 之间互转。

现在这个锁被转到 9–0–4，如图所示，而她的开锁密码为 5–8–7。



请问她最少要转动多少次才能转到她的开锁密码？

Felicity has a combination lock for her bike like the one below. It has the numbers 0 to 9 on each tumbler.

It clicks every time she moves the tumblers one number forward or back, including a click as the tumbler moves between 9 and 0.

She found the lock in the position 9–0–4 shown. Her combination is 5–8–7.

What is the *least* number of clicks needed to get the lock to her combination?

(A) 20　(B) 18　(C) 17　(D) 9　(E) 7

10. 请问下列哪一个选项内的数与自己相乘之乘积恰等于 5 乘以 20？

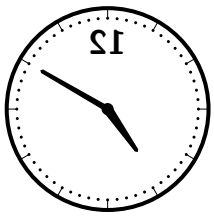
Which number multiplied by itself is equal to 5 times 20?

(A) 10 (B) 20 (C) 25 (D) 100 (E) 120

11-20 题，每题 4 分
Questions 11 to 20, 4 marks each

11. 小桂在镜子中看到一个时钟的样子如图所示。请问它的准确时间是什么？

Greg sees a clock in the mirror, where it looks like this. What is the actual time?



(A) 4:10 (B) 4:50 (C) 5:10 (D) 6:50 (E) 7:10

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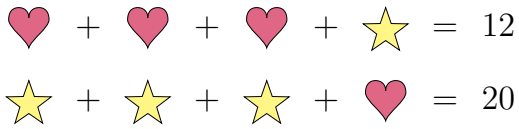
12. 已知以下两条算式



请问 ♥ 所代表的值是多少？

In these two number sentences

what is the value of ♥ ?

(A) 1 (B) 2 (C) 3 (D) 4 (E) 5

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13. 在如图的加式中，字母 X、Y 与 Z 都分别代表不同的数码。请问字母 X 所代表的数码是什么？

In this sum, each of the letters X, Y and Z represents a different digit. Which digit does the letter X represent?

$$\begin{array}{r} X\ X \\ +\ X\ Y \\ \hline Y\ X\ Z \end{array}$$

(A) 1 (B) 4 (C) 5 (D) 7 (E) 9

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14. 一个爱好数学的学生造了如下所示的表。

在此表上每一行两端的数都逐行往下增加 1 而其余的数都是将它上方的两个数相加而得。

依此规则，请问接下来的一行上所有数之总和是多少？

A maths student made the following pattern:

1	1	1				
2	2	2	2			
3	4	4	4	3		
4	7	8	8	7	4	
5	11	15	16	15	11	5

The numbers down the sides of the pattern increase by 1 and each of the other numbers is found by adding the two numbers above it.

What will be the sum of all the numbers on the next line in this pattern?

(A) 128 (B) 138 (C) 148 (D) 158 (E) 168

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15. 学校用 \$900 买了 18 箱的油漆。已知每一箱内都有相同瓶数的油漆，且每瓶的价格为 \$2.50，请问每一箱内各有多少瓶油漆？

The school bought 18 boxes of primary school paint for \$900. Each box had a number of bottles, each worth \$2.50. How many bottles were in each box?

(A) 15 (B) 20 (C) 45 (D) 50 (E) 125

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16. 在某年六月，共有四个星期三与五个星期二。请问这年的六月一日是星期几？

(A) 星期一 (B) 星期二 (C) 星期四 (D) 星期五 (E) 星期六

One year in June, there were four Wednesdays and five Tuesdays. On which day was the first of June?

(A) Monday (B) Tuesday (C) Thursday (D) Friday (E) Saturday

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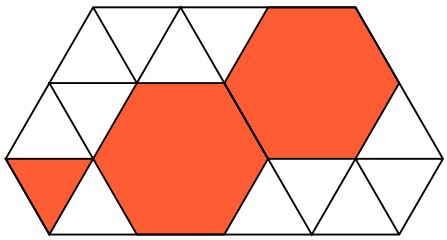
17. 请问图中有百分之几被涂上阴影？

What percentage of this shape is shaded?



(A) 40%　(B) 48%　(C) 50%　(D) 52%　(E) 66%

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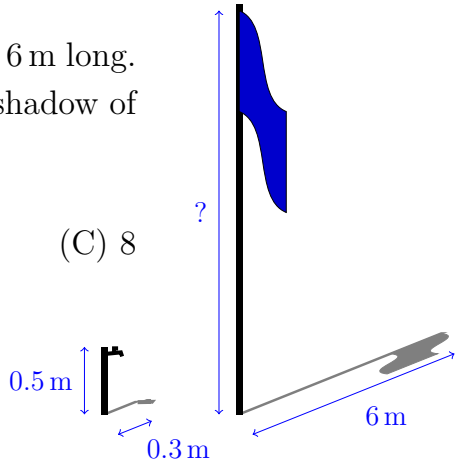
18. 在上午 10 点时，学校旗杆的阴影长度为 6 m。
旗杆旁边另有一高度为 0.5 m 的水管且其阴影长度为 0.3 m。
请问学校旗杆的高度为多少 m？

At 10 am the school flagpole cast a shadow 6 m long.
Next to the flagpole, the 0.5 m tap cast a shadow of 0.3 m.
How tall is the flagpole in metres?

(A) 3　(B) 5　(C) 8　(D) 10　(E) 12



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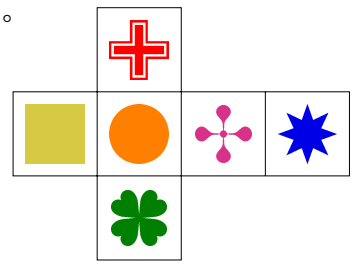
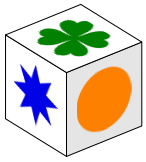
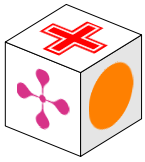
19. 可将如右图所示的图案折成一个正立方体。
请问所折出来的正立方体是哪一个？

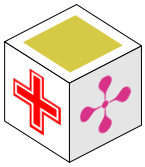
This shape can be folded up to make a cube.
Which cube could it make?

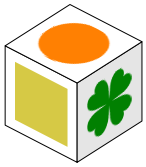


(A)


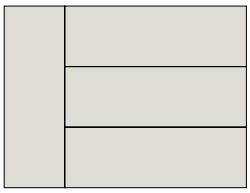
(B)

(C)


(D)


(E)


20. 已知大长方形的面积为 300 m^2。这个大长方形可以切成四个全等的小长方形，如图所示。请问小长方形的宽为多少 m？



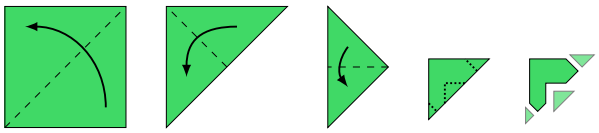
The area of the large rectangle is 300 square metres. It is made up of four identical smaller rectangles. What is the width of one of the small rectangles in metres?

(A) 2 (B) 5 (C) 8 (D) 10 (E) 12

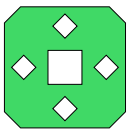
21-25 题，每题 5 分
Questions 21 to 25, 5 marks each

21. 将一张纸经过如下图所示的方式折迭与剪裁，请问展开后会得到哪一个选项的图案？

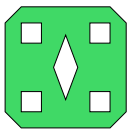
Which one of the patterns below would be created with these folds and cuts?



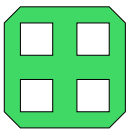
(A)



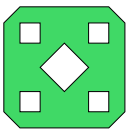
(B)



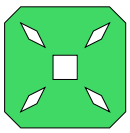
(C)



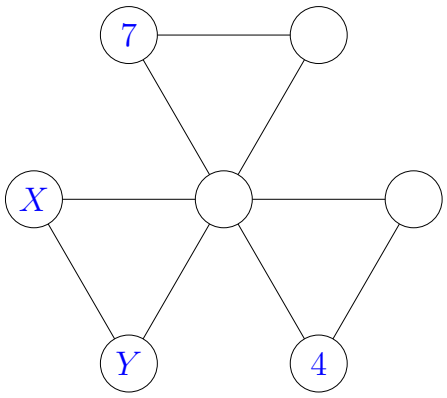
(D)



(E)



22. 将正整数 1、2、3、4、5、6、7 不重复地恰填入图中的七个圆圈内，使得所画出的三个三角形顶点上的数之和都相等。
现已填入其中两个数，如图所示。
请问 $X+Y$ 的值是多少？



The whole numbers from 1 to 7 are to be placed in the seven circles in the diagram. In each of the three triangles drawn, the sum of the three numbers is the same.
Two of the numbers are given.
What is $X+Y$?

(A) 5 (B) 6 (C) 7 (D) 8 (E) 9

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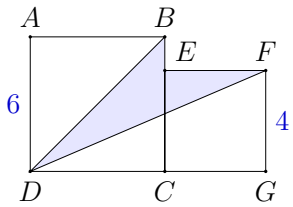
23. 将一个边长为 6 cm 的正方形 $ABCD$ 与一个边长为 4 cm 的正方形 $EFGC$ 并在一起，如图所示。
请问阴影区域 $BDFE$ 的面积为多少？



A square $ABCD$ with a side of 6 cm is joined with a smaller square $EFGC$ with a side of 4 cm as shown.
What is the area of the shaded shape $BDFE$?

(A) $12\,\text{cm}^2$ (B) $14\,\text{cm}^2$ (C) $16\,\text{cm}^2$ (D) $18\,\text{cm}^2$ (E) $24\,\text{cm}^2$

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24. 在 2017 这一年，我的家庭非常奇特：我 7 岁、我的弟弟 5 岁、我的妈妈 29 岁而我的爸爸 31 岁，我们的岁数都是质数。
下一次我的家人再度这样奇特，即我们的岁数再度都为质数时，请问我的爸爸之年龄为几岁？

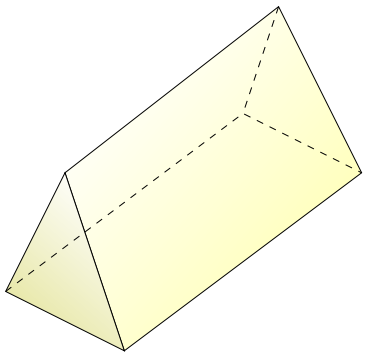
In this year of 2017, my family is *in its prime*: I am 7, my brother is 5, my mother is 29 and my father is 31. All of our ages are prime numbers.
What is my father's age the next year that my family is *in its prime*, when all of our ages are again prime?

(A) 37 (B) 41 (C) 43 (D) 47 (E) 61

25. 将一个三角体柱用一个平面切一次而分成两片。请问这两片的面数之总和的最小可能值是多少？

A triangular prism is to be cut into two pieces with a single straight cut. What is the smallest possible total for the combined number of faces of the two pieces?

(A) 6 (B) 8 (C) 9 (D) 10 (E) 11



问题 26-30 的答案为 000-999 之间的整数，请将答案填在答案卡上对应的位置。

第 26 题占 6 分，第 27 题占 7 分，第 28 题占 8 分，第 29 题占 9 分，第 30 题占 10 分。

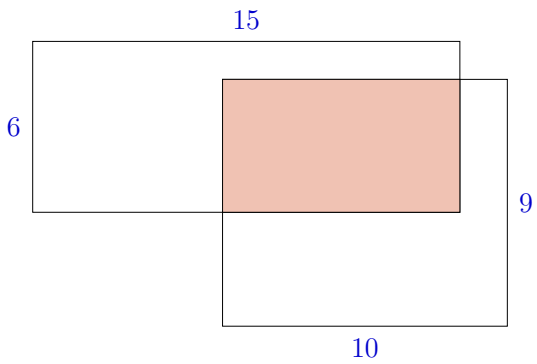
For questions 26 to 30, shade the answer as a whole number from 0 to 999 in the space provided on the answer sheet.

Question 26 is 6 marks, question 27 is 7 marks, question 28 is 8 marks, question 29 is 9 marks and question 30 is 10 marks.

26. 将两个矩形重迭造成三个面积相等的区域，原来的两个矩形分别是 6 cm × 15 cm 与 10 cm × 9 cm，如图所示。若阴影区域小矩形的每一边之长度都是整数 cm，请问阴影区域小矩形的周长为多少 cm？

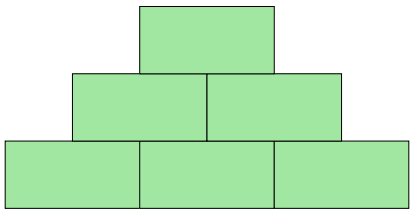
Two rectangles overlap to create three regions, each of equal area. The original rectangles are 6 cm by 15 cm and 10 cm by 9 cm as shown. The sides of the smaller shaded rectangle are each a whole number of centimetres.

What is the perimeter of the smaller shaded rectangle, in centimetres?



27. 小强利用一些长为 2 cm、宽为 1 cm 的长方形纸牌构造一个塔，其中每一列都比上方的列多一张卡片。
已知三层的塔之周长为 18 cm，如图所示。
请问十层的塔之周长为多少 cm？

Jonathan made a tower with rectangular cards 2 cm long and 1 cm wide, where each row has one more card than the row above it.
The perimeter of a tower with 3 levels is 18 cm, as shown.
What will be the perimeter of a tower with 10 levels, in centimetres?

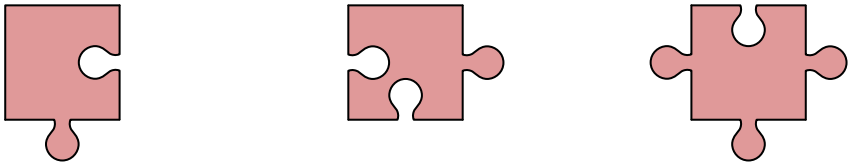


28. 从 0 到 9 的十个数码都各恰使用一次可构成两个五位数。请问这两个五位数之差的最小可能值是多少？

All of the digits from 0 to 9 are used to form two 5-digit numbers. What is the smallest possible difference between these two numbers?

29. 一块拼图的配件是由一张正方形硬纸板，在它的至少二条边上分别各有一个「插榫」或「接榫」的组合。

拼图配件分别有角落、棱边或内部的配件，如下图所示。



角落的配件
(两条直线边构成一直角)

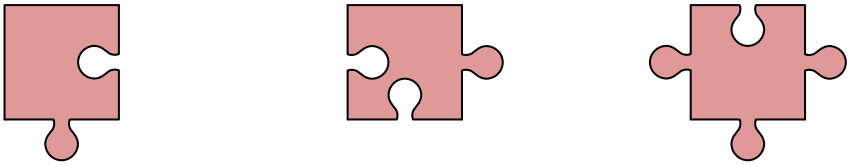
棱边的配件
(有一条直线边)

内部的配件
(没有直线边)

若一块拼图的配件经过旋转 (不可以翻转) 后会与另外一块重合，则认定这两块配件的形状是相同的。请问总共有多少种不同可能形状的拼图配件？

A jigsaw piece is formed from a square with a combination of 'tabs' and 'slots' on at least two of its sides.

Pieces are either corner, edge or interior, as shown.



corner piece
(two straight sides at right angles)

edge piece
(one straight side)

interior piece
(no straight sides)

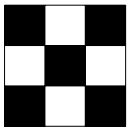
We treat two shapes as the same if one is a rotation of the other, without turning it over. How many different shapes are possible?

30. 一个 3×3 方格表的小方格内都被涂上了黑色或是白色。

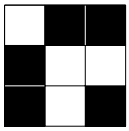
若其中任何一个 2×2 子方格表内每一种颜色都恰各占有二个小方格，则称这个方格表为均衡的，如第一个范例。

第二个范例则是不均衡的，因为右下角的 2×2 子方格表内有三个白色小方格。

若将旋转或翻转后不同的方格表视为不同，请问总共有多少个均衡的 3×3 方格表？



均衡的
balanced



不均衡的
unbalanced

A 3×3 grid has a pattern of black and white squares.

A pattern is called *balanced* if each 2×2 subgrid contains exactly two squares of each colour, as seen in the first example.

The pattern in the second example is *unbalanced* because the bottom-right 2×2 subgrid contains three white squares.

Counting rotations and reflections as different, how many balanced 3×3 patterns are there?

Solutions – Upper Primary Division

1. $17 + 20 = 37$,

hence (C).

2. There are 5 per kilogram, so 20,

hence (B).

3. (Also MP6)
Three out of the five dice showed a two on top,

hence (E).

4. She spent $55 + 20 = \$75$, so her change is $100 - 75 = \$25$,

hence (A).

5. (Also MP9)
Only shapes 1 and 3 have five sides and therefore are pentagons,

hence (D).

6. Naomi lives $3 \times 4 = 12$ km from school.
Olivia lives $12 - 3 = 9$ km from school,

hence (A).

7. (Also MP12)
Subtract the 73 and then add 37 to get the correct answer: $157 - 73 + 37 = 121$,

hence (B).

8. (Also MP13)
The squares and triangles are worth $4 \times 6 + 3 \times 6 = 24 + 18 = 42$ cents. Then the hexagon is worth $50 - 42 = 8$ cents,

hence (B).

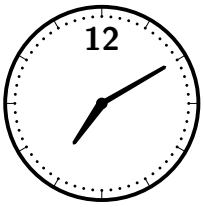
9. Tumbler 1: 9 to 5 is 4 clicks back.
Tumbler 2: 0 to 8 is 2 clicks back.
Tumbler 3: 4 to 7 is 3 clicks forward.
Total clicks are 9,

hence (D).

10. The number multiplied by itself equals 100. Since $10 \times 10 = 100$, the number is 10,

hence (A).

11. (Also MP15)
Reflect the clock face in a vertical axis so that it is the right way around.



Then the time is 10 minutes past 7,

hence (E).

12. (Also MP17)
Alternative 1
Adding both left-hand sides, 4 hearts and 4 stars are equal to 32, so ☆ + ♥ = 8.
Then in the first line, ♥ + ♥ + 8 = 12 so that ♥ + ♥ = 4 and ♥ = 2,

hence (B).

Alternative 2
We can make this pattern.

♥	+	♥	+	♥	+	♥	=	☐
♥	+	♥	+	♥	+	☆	=	12
♥	+	♥	+	☆	+	☆	=	☐
♥	+	☆	+	☆	+	☆	=	20
☆	+	☆	+	☆	+	☆	=	☐

Each line replaces one ♥ with one ☆, so the numbers ☐, 12, ☐, 20, ☐ will be equally spaced.
This means that they are 4 apart: $\boxed{8}$, 12, $\boxed{16}$, 20, $\boxed{24}$. Then ♥ = 8 ÷ 4 = 2,

hence (B).

13. In the tens column, $X + X = X$. This can only occur if $X = 0$ or if $X = 9$ and there is a carry from the ones column.
However, if $X = 0$ then the ones column has $Y = Z$, which is not allowed.
If $X = 9$, then $Y = 1$ and then the sum is $99 + 91 = 190$,

hence (E).

14. If we add up each row we get 3, 8, 18, 38 and 78. Each number is 2 times the previous number plus 2. So the next row adds to $78 \times 2 + 2 = 158$,

hence (D).

Note. The double-plus-two rule for each row total works because every number in the pattern contributes to the two numbers below it, with an extra +1 contributed at each end:

$$\begin{array}{ccccccccccccc} & 4 & & 7 & & 8 & & 8 & & 7 & & 4 & \\ \swarrow & & \searrow & & \swarrow & & \searrow & & \swarrow & & \searrow & & \swarrow & & \searrow \\ 1{+}4 & & 4{+}7 & & 7{+}8 & & 8{+}8 & & 8{+}7 & & 7{+}4 & & 4{+}1 \end{array}$$

15. Each box is worth $\$900 \div 18 = \$50$.

Each bottle is worth \$2.50, so 4 bottles are worth \$10. Then each \$50 box will have $5 \times 4 = 20$ bottles,

hence (B).

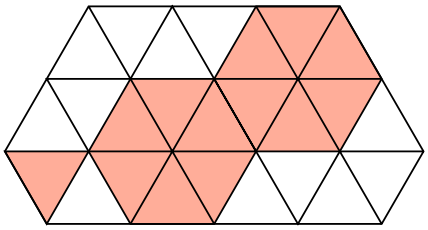
16. (Also MP18)

June has 30 days, which is 4 weeks and 2 days. So the last two days of June are the same day of the week as the first two. Thus these two days are the only ones that appear 5 times.

Now there are 5 Tuesdays, but not 5 Wednesdays, so there must be 5 Mondays. That is, the first two days and the last two days are Mondays and Tuesdays,

hence (A).

17. Subdivide both regular hexagons into 6 equilateral triangles. This gives a grid of 25 identical equilateral triangles, with 13 shaded:



As a percentage of the shape, each triangle is $100\% \div 25 = 4\%$ and so the 13 shaded triangles are $13 \times 4\% = 52\%$,

hence (D).

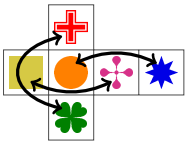
18. The flagpole's shadow is 20 times longer than the tap's shadow. Then the flagpole is 20 times taller than the tap.

So the flagpole's height is $20 \times 0.5\,\text{m} = 10\,\text{m}$,

hence (D).

19. (Also MP21)

In the net we can find the opposite faces.



- is opposite which rules out (D).
- is opposite which rules out (A).
- is opposite which rules out (B).

We can also rule out (C) by looking at what happens if is the top face of the cube and and are visible. This puts on the left and on the right.

The cube in (E) is possible, since with on top, and with on the left, is on the right,

hence (E).

20. The long side of one small rectangle matches 3 short sides of the other small rectangles. So the small rectangles can each be cut into 3 squares. This cuts the large rectangle into 12 squares, each of area $300 \div 12 = 25$ square metres.

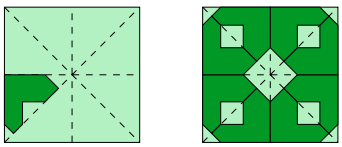
These squares will be 5 metres by 5 metres, and so the small rectangles are 15 metres long and 5 metres wide,

hence (B).

21. (Also MP23)

The folded paper has 8 layers when it is cut.

One of these cut layers is shown on the original square on the left, and all 8 are shown on the right,



hence (D).

22. (Also J18)

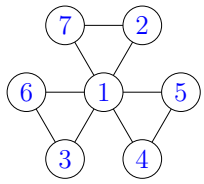
Alternative 1

Let C be the number in the centre, so that the sum on each triangle is $T = X+Y+C$. Adding all three triangles gives $3T$, and this includes all 7 numbers, except that the centre number is counted three times. That is,

$$3T = 1+2+3+4+5+6+7+2C = 28+2C$$

For $28+2C$ to be a multiple of 3, $C = 1$, 4 or 7. However 4 and 7 are already in the diagram, so that $C = 1$.

Then $3T = 28+2 = 30$, which means that $T = 10$ and $X+Y = 9$. The diagram on the right shows one of the two possible solutions with $X+Y = 9$,



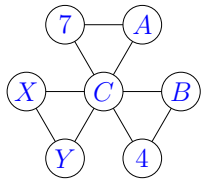
hence (E).

Alternative 2

Label the diagram as shown and let $Q = X+Y$. Each triangle has the same total, so $Q = X+Y = 7+A = B+4$. Clearly $Q \geqslant 8$.

If $Q = 8$, then $B = 4$, which would repeat 4. So $Q \neq 8$.

If $Q = 9$, then $A = 2$ and $B = 5$. The remaining three numbers are 1, 3 and 6, so that X and Y are 3 and 6 (in either order) and $C = 1$. That is, there are two solutions when $Q = 9$.



If $Q = 10$, then $A = 3$ and $B = 6$. The remaining numbers are 1, 2 and 4. However, $X+Y = 10$, so that $Q = 10$ is not possible.

If $Q > 10$, then $B > 6$, which is not possible.

Consequently there are only two solutions, both with $X+Y = 3+6 = 9$,

hence (E).

23. The combined area of squares $ABCD$ and $EFGC$ is:

$$6^2 + 4^2 = 36 + 16 = 52\,\text{cm}^2$$

The combined area of the unshaded triangles $\triangle ABD$ and $\triangle DFG$ is:

$$\frac{1}{2} \times 6 \times 6 + \frac{1}{2} \times 10 \times 4 = 18 + 20 = 38\,\text{cm}^2$$

Therefore the shaded area is $52 - 38 = 14\,\text{cm}^2$,

hence (B).

24. Since no even numbers other than 2 will be prime, we just have to keep adding 2 to all the ages and check whether the result gives 4 primes.

2017	5✓	7✓	29✓	31✓
2019	7✓	9	31✓	33
2021	9	11✓	33	35
2023	11✓	13✓	35	37✓
2025	13✓	15	37✓	39
2027	15	17✓	39	41✓
2029	17✓	19✓	41✓	43✓

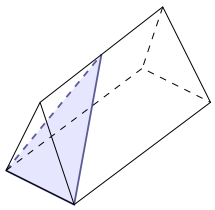
This occurs in 2029 when my father will be 43,

hence (C).

Comment As well as the ages, the years 2017 and 2029 are also primes.

25. (Also J23)

The original prism has 5 faces. The cut will expose two totally new faces (either side of the cut) and any face it passes through will become two faces. So there are at least 7 faces in the final two polyhedra, and the minimum combined number of faces will occur when the cut passes through as few faces as possible.



The diagram shows that a cut along one edge and passing through two faces is possible, creating two pieces with a total of 9 faces.

If 7 faces were the minimum, they would be the two exposed by the cut and the 5 original faces. The plane would cut the surface of the polyhedron only through its edges. However there is no such plane: once a plane contains two or more edges of the prism, it is a face of the prism.

If 8 faces were the minimum, these would be the two faces exposed by the cut, 4 faces that are not cut and one face that is cut in two. Again, such a cut would contain two edges of the prism, so it is not possible.

Consequently the minimum number of faces in the two pieces is 9,

hence (C).

26. (Also J21)
The 15×6 rectangle has area $90\,\text{cm}^2$, and the shaded area is half of this, which is $45\,\text{cm}^2$. This is also half of the 10×9 rectangle.

This shaded rectangle can then be 45×1, 15×3 or 9×5, but only the 9×5 rectangle will fit inside the 15×6 rectangle as shown. Consequently the perimeter of the shaded rectangle is $2 \times (5 + 9) = 28\,\text{cm}$,

hence (28).

27. The shape will be 20 cm wide and 10 cm tall. So the total length of the left-facing edges will be 10 cm.

Similarly the total length of the right-facing edges is 10 cm, the length of the upward-facing edges is 20 cm and the length of the bottom edge is 20 cm.

In all, the perimeter is $10 + 10 + 20 + 20 = 60\,\text{cm}$,

hence (60).

28. (Also MP30, J26, I19, S18)
The numbers must have different first digits, and so the smallest possible difference will be when the first digits differ by 1. Furthermore, the number formed by the remaining digits of the smaller number will be as large as possible and the number formed by the remaining digits of the larger number will be as small as possible.

The largest and smallest possible numbers formed from 4 digits are 9876 and 0123, so the two 5-digit numbers are 50123 and 49876 and their difference is $50123 - 49876 = 247$,

hence (247).

29. (Also J27)
Denoting straight edges by 'E', slots by 'S' and tabs by 'T', each shape can be represented by a string of four of these symbols, subject to the following rules:

- There can be at most two E's.
- If there are two E's, they are adjacent.
- Cyclic permutations are considered equal: ESST = TESS = STES = SSTE.

The possible pieces are listed in the following table, where $\star$ represents 'T' or 'S':

Type	**Form**	**Number**	**List**
Corner piece	E E $\star\star$	$2^2 = 4$	EESS EETS EEST EETT
Edge piece	E $\star\star\star$	$2^3 = 8$	ESSS ETSS ESST ETST ESTS ETTS ESTT ETTT
Interior piece	$\star\star\star\star$	6	SSSS STST SSST STTT SSTT TTTT
Total		**18**	

There are 18 such pieces in total,

hence (18).

30. (Also J25)

Alternative 1

For each balanced pattern there is another which can be obtained by switching black and white throughout the entire grid. Hence we may assume the top-left square is black and double the number of such patterns. The top-left 2×2 subgrid can be coloured in three ways, with the grey squares yet to be determined:

(1) (2) (3)

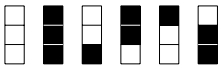
In case (1), a and b must both be black so the top-right 2×2 subgrid is balanced; the bottom row can alternate either black–white–black or white–black–white, hence there are 2 patterns for case (1). Since case (2) is the reflection of case (1) along a diagonal, there are also 2 patterns for case (2).

Case (3) allows two possibilities for a and b: if a is black and b is white then the bottom row can again alternate in either of two ways, and if a is white and b is black then the bottom row must be black–white–white, hence there are 3 patterns for case (3). This gives 7 patterns in total when the top-left square is black, so doubling gives a total of 14 patterns,

hence (14).

Alternative 2

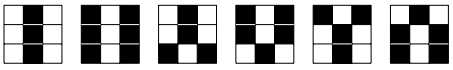
Consider the pattern as made up of 3 columns. There are 8 possibilities for the first column, which can either be non-alternating or alternating.



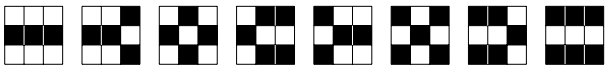
non-alternating

alternating

In the six non-alternating possibilities for the first column, there are two (or more) adjacent squares in the column that are the same colour. To be balanced, the corresponding squares in the next column must be the reversed colour. Then every square in the second column is forced to be the reverse of the corresponding square in the first column. For the same reason, the third column must be the reversal of the second column. Then each row alternates black–white–black or white–black–white. So there are 6 patterns where the first column is non-alternating:



In the two alternating possibilities for the first column, the next column must also be alternating, but there are two choices—it can either be the same or reversed. Likewise there are two choices for the third column. In all, there are 8 balanced patterns where the first column is alternating:



Combining these two cases, the number of balanced patterns is $6 + 8 = 14$,

hence (14).

Answers

Question	Middle Primary	Upper Primary	Junior	Intermediate	Senior
1	A	C	A	C	A
2	C	B	B	A	A
3	E	E	A	C	C
4	A	A	B	C	D
5	D	D	D	C	D
6	E	A	A	E	B
7	A	B	D	B	C
8	C	B	D	A	E
9	D	D	C	A	C
10	D	A	E	D	D
11	A	E	B	E	C
12	B	B	D	C	E
13	B	E	E	E	E
14	C	D	A	D	B
15	E	B	C	D	C
16	C	A	A	B	B
17	B	D	D	B	D
18	A	D	E	E	D
19	E	E	E	D	A
20	C	B	D	E	B
21	E	D	B	C	E
22	C	E	B	A	B
23	D	B	C	D	C
24	C	C	B	E	D
25	D	C	D	E	E
26	337	28	247	807	13
27	13	60	18	648	234
28	4	247	27	56	56
29	468	18	286	286	360
30	247	14	234	99	510